



SAPIENZA
UNIVERSITÀ DI ROMA

“SAPIENZA” UNIVERSITY OF ROME

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COMPUTER SCIENCE AND STATISTICS

DEPARTMENT OF COMPUTER, CONTROL
AND MANAGEMENT ENGINEERING

MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE AND ROBOTICS

Reinforcement Learning

Lecture notes

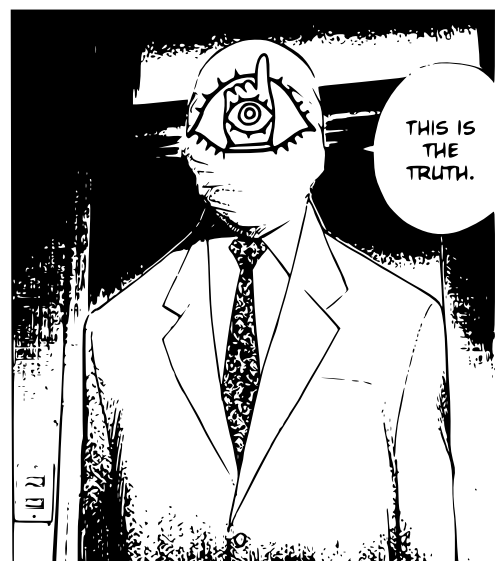
Author

Marco Casu

<https://github.com/CasuFrost>

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This document summarizes and presents the topics for the Reinforcement Learning course for the Master's degree in Artificial Intelligence and Robotics at Sapienza University of Rome. The document is free for any use. If the reader notices any typos, they are kindly requested to report them to the author.



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CHAPTER

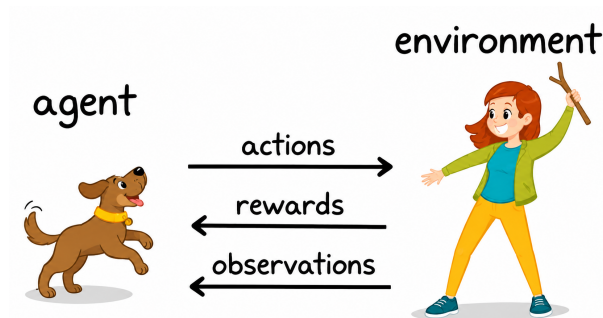
1

EXPLORATION AND EXPLOITATION

Definition 1.0.1 : Reinforcement Learning

With **Reinforcement Learning** we mean the process to *learn* what to do in order to maximize a numerical reward signal:

- The learner must discover which actions yield the highest reward by trying them.
- The actions might affect not only immediate reward, but also subsequent ones. It is a *trial-and-error* process.



The agent has to:

- **Exploit** what it has already experienced to obtain a reward.
- **Explore** in order to make better action selections in the future.

One of the main challenges of reinforcement learning is the exploration-exploitation trade-off. In stochastic tasks, each action must be tried many times, in order to correctly estimate the expected reward.

1.1 Multi-Armed Bandits

One of the main problems in reinforcement learning is to decide when we should continue to explore (collect more information about the environment) or to exploit (use the available information in order to

get closer to the goal). We need to *carefully* and *systematically* explore. The **multi-armed bandits (MAB)** problem is a simplified problem that describes the exploration paradigm.

We have K different arms, that are our actions:

$$a_1, a_2, \dots, a_K \quad (1.1)$$

and a reward $r \in [0, 1]$, that is the scalar signal representing a feedback. Each single arm has an unknown reward distribution v_i with mean μ_i , every time we pull one of the K arms we observe a reward. We interact with the MAB in the following way:

- for $t = 0, 1, \dots, T - 1$:
 1. pull an arm $I_t \in \{a_1, a_2, \dots, a_K\}$ based on historical information
 2. observe the reward $r_t \sim v_{I_t}$ of the arm I_t .

The goal of this process is to minimize the **opportunity loss**. Assume that the best arm to pull is known (a^*), with its mean reward (μ^*):

$$\mu^* = \max_{i \in \{1, \dots, K\}} \mu_i. \quad (1.2)$$

Consider the total expected reward obtained if we pull the best arm for T rounds:

$$T\mu^* \quad (1.3)$$

and the total expected reward of the arms we actually pulled over T rounds:

$$\sum_{t=0}^{T-1} \mu_{I_t} \quad (1.4)$$

the **regret** is the difference between these two quantities:

$$\text{Regret}_T = T\mu^* - \sum_{t=0}^{T-1} \mu_{I_t} \quad (1.5)$$

the regret is the expression of the opportunity loss, which we want to minimize. In the MAB example, the exploration-exploitation trade-off manifests itself in the form of the following question:

Should we pull arms that have been tried less frequently in the past (i.e., explore), or should we commit to the current best arm (i.e., exploit)?

We consider different strategies. The **greedy algorithm** tries each arm once, and then commits to the one that has the highest observed reward. This algorithm is not optimal: a low mean arm might generate a high reward the single time we sample it.

1.1.1 The Hoeffding Inequality and Regret Calculation

A better choice of algorithm is the **Explore & Commit**: we try each arm multiple times, and we compute the empirical mean of the rewards. Then we commit to the arm with the highest empirical mean.

1. Let N be a fixed value, and $T \geq NK$
2. for $k = 1, \dots, K$
 - pull the arm a_k for N times
 - observe the set of observations $\{r_1^{(k)}, r_2^{(k)}, \dots, r_N^{(k)}\}$
 - compute the empirical mean $\hat{\mu}_k = \frac{1}{N} \sum_{i=1}^N r_i^{(k)}$
3. for $t = NK, \dots, T - 1$

- pull the best empirical arm $I_t = \arg \max_{i \in \{1, \dots, K\}} \hat{\mu}_i$

Proposition 1.1.1 : Hoeffding Inequality

Given a distribution with mean μ , where we have observed N independent and bounded samples $r_i \in [0, 1]$, the following inequality holds with probability $1 - \delta$:

$$\left| \frac{1}{N} \sum_{i=1}^N r_i - \mu \right| \leq \sqrt{\frac{\ln(2/\delta)}{2N}}. \quad (1.6)$$

The Proposition 1.1.1 gives us a confidence interval on our empirical mean.

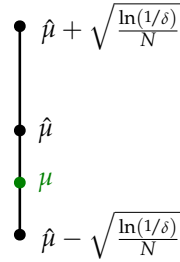


Figure 1.1: The true mean is located at a bounded distance from the sampled mean with probability $1 - \delta$.

For example, if $\delta = 0.01$, the bound holds with probability 99%. The Hoeffding inequality can also be applied for all the arms at once. During the exploration phase, for **all the arms**: with probability $1 - \delta$ the following inequalities hold:

$$\left| \frac{1}{N} \sum_{i=1}^N r_i^{(k)} - \mu_k \right| \leq \sqrt{\frac{\ln(2K/\delta)}{2N}} \quad \text{for every } k \in \{1, \dots, K\}. \quad (1.7)$$

This comes from a union bound over the K arms, which is why $2/\delta$ becomes $\frac{2K}{\delta}$.

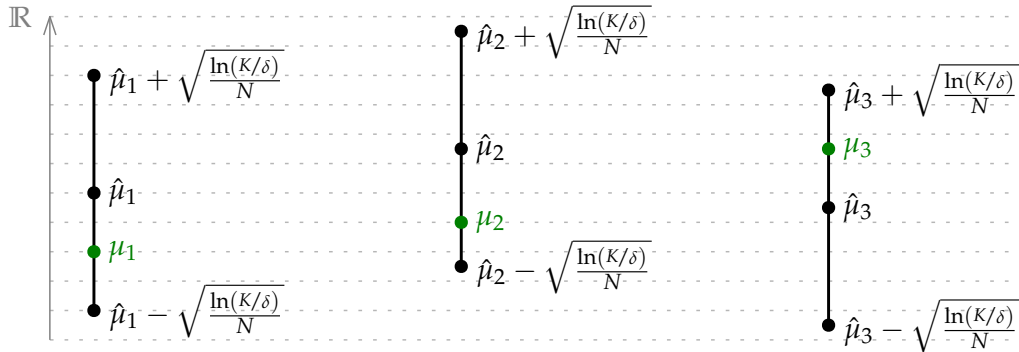


Figure 1.2: Hoeffding inequalities for multiple arms.

In Figure 1.2 qualitatively shows an example of how the Explore & Commit algorithm would fail: in this case we would choose the arm a_2 as the sample mean $\hat{\mu}_2$ is higher than the sample mean of the other arms, but we can clearly see how the true mean μ_3 of arm a_3 is higher than the true mean μ_2 of the arm a_2 . If we choose the arm a_2 , the loss of accumulated reward due to this suboptimal decision is called **regret** and is equal to:

$$\text{Regret}_{\text{exploit}} = (T - NK)(\mu_3 - \mu_2) \quad (1.8)$$

where $T - NK$ is the number of remaining steps dedicated to exploitation (total steps T minus those used to explore NK). The difference $(\mu_3 - \mu_2)$ is small, less than the confidence ranges.

Since we assumed that the reward range is $[0, 1]$, the maximum loss for a single round is at most 1, so the regret accumulated in the exploration phase is bounded above by:

$$\text{Regret}_{\text{explore}} \leq N \left(\underbrace{K-1}_{\text{one is optimal}} \right) \leq NK. \quad (1.9)$$

Let $\hat{I}$ be the empirical best arm and I^* the true best arm, the worst possible regret in exploitation is

$$\text{Regret}_{\text{exploit}} \leq (T - NK)(\mu_{I^*} - \mu_{\hat{I}}) \quad (1.10)$$

where

$$\mu_{I^*} - \mu_{\hat{I}} \leq \left[\hat{\mu}_{I^*} + \sqrt{\frac{\ln(K/\delta)}{N}} \right] - \left[\hat{\mu}_{\hat{I}} - \sqrt{\frac{\ln(K/\delta)}{N}} \right] \quad (1.11)$$

$$= \underbrace{\hat{\mu}_{I^*} - \hat{\mu}_{\hat{I}}}_{\leq 0} + 2\sqrt{\frac{\ln(K/\delta)}{N}} \leq 2\sqrt{\frac{\ln(K/\delta)}{N}} \quad (1.12)$$

hence the regret is at most twice our confidence interval:

$$\text{Regret}_{\text{exploit}} \leq 2T\sqrt{\frac{\ln(K/\delta)}{N}}. \quad (1.13)$$

The total regret is

$$\text{Regret}_T = \text{Regret}_{\text{explore}} + \text{Regret}_{\text{exploit}} \leq NK + 2T\sqrt{\frac{\ln(K/\delta)}{N}}. \quad (1.14)$$

We want to minimize the regret by choosing the optimal value of N , this can be done by taking the derivative with respect to N of the regret and setting it to zero, then we solve for N . This leads to:

$$N = \left(\frac{T\sqrt{\ln(K/\delta)}}{K} \right)^{\frac{2}{3}} \quad (1.15)$$

The following inequality holds

$$\text{Regret}_T = O(T^{\frac{2}{3}} K^{\frac{1}{3}} \ln^{\frac{1}{3}}(K/\delta)). \quad (1.16)$$

This **growth rate** of the regret using the Explore & Commit algorithm is kind of slow. We would like to get something faster, like $O(\sqrt{T})$ (that is proven to be the minimum we can get). We want to design a new algorithm, first of all we need a list of generic statistics useful to maintain in order to compute our confidence bounds and the regret:

- we denote $N_t(i)$ the number of times we have tried the arm a_i so far:

$$N_t(i) = \sum_{\tau=0}^{t-1} \mathbb{1}\{I_\tau = a_i\} \quad (1.17)$$

- the empirical mean so far:

$$\hat{\mu}_t(i) = \sum_{\tau=0}^{t-1} \mathbb{1}\{I_\tau = a_i\} \frac{r_\tau}{N_t(i)} \quad (1.18)$$

- the confidence with probability $1 - \delta$:

$$|\hat{\mu}_t(i) - \mu_i| \leq \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}. \quad (1.19)$$

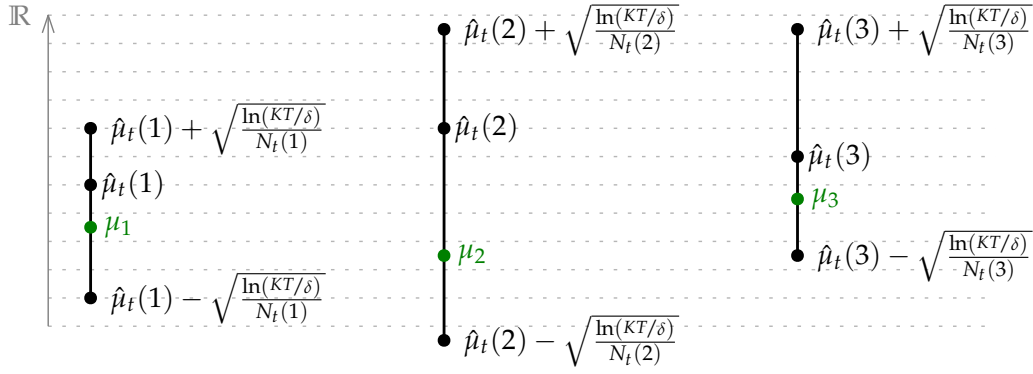


Figure 1.3: In the confidence interval in Equation (1.19), the length depends on how many times one arm has been tried.

For example, in Figure 1.3 the confidence interval of arm 2 is wider because we did not try arm 2 as many times as arm 1 and 3. We select the arm with the highest upper confidence bound, and we make an optimistic assumption: the true mean actually corresponds to the upper confidence bound.

- If the bound is large, it means either that arm has been tested few times, therefore if that is not good, the more we choose it the more the upper bound decreases and eventually another one will be the highest, or that it is actually good.
- If a confidence interval is large then we want to explore, a small interval on the selected arm means its high UCB reflects a genuinely high mean.

The **UCB algorithm** works as follows:

- For the first K iterations, we pull each arm once
- for $t = K, \dots, T - 1$:
 - We pick the arm with the highest upper confidence bound:

$$I_t = \arg \max_{i=1, \dots, K} \left(\hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}} \right) \quad (1.20)$$

- update the statistics.

Now let's compute the regret at a given time step t , we have:

$$\mu^* - \mu_{I_t} \leq \hat{\mu}_t(I_t) + \sqrt{\frac{\ln(KT/\delta)}{N_t(I_t)}} - \mu_{I_t} \leq 2\sqrt{\frac{\ln(KT/\delta)}{N_t(I_t)}} \quad (1.21)$$

if N_t is small, we have a high regret but we have to explore, if it's large, we exploit and the regret is small. The total regret summing over all iterations is:

$$\text{Regret}_T = \sum_{t=0}^{T-1} (\mu^* - \mu_{I_t}) \leq \sum_{t=0}^{T-1} 2\sqrt{\frac{\ln(KT/\delta)}{N_t(I_t)}} \leq 2\sqrt{\ln(KT/\delta)} \cdot \underbrace{\sum_{t=0}^{T-1} \sqrt{\frac{1}{N_t(I_t)}}}_{\leq O(\sqrt{KT})}. \quad (1.22)$$

Theorem 1.1.1 : Regret of the UCB

With the UCB algorithm, the following bound holds:

$$\text{Regret}_T = O\left(\sqrt{KT \ln\left(\frac{KT}{\delta}\right)}\right) \quad (1.23)$$

or

$$\text{Regret}_T = \tilde{O}(\sqrt{KT}) \quad (1.24)$$

with probability $1 - \delta$.

Proof: We want to bound the summation term appearing in the UCB regret bound. First we cluster the pulled arms into K groups, where the i -th group contains all steps where arm i is pulled:

$$\sum_{t=0}^{T-1} \sqrt{\frac{1}{n_t(I_t)}} = \sum_{i=1}^K \sum_{t=0}^{T-1} \mathbb{1}\{I_t = a_i\} \sqrt{\frac{1}{n_t(i)}} \quad (1.25)$$

where $n_t(i)$ is the size of the specific group.

$$\sum_{i=1}^K \sum_{t=1}^{T-1} \mathbb{1}\{I_t = a_i\} \sqrt{\frac{1}{n_t(i)}} = \sum_{i=1}^K \sum_{t=1}^{n_T(i)} \sqrt{\frac{1}{t}}. \quad (1.26)$$

We use the following inequality:

$$\sum_{k=1}^N \frac{1}{\sqrt{k}} \leq 2\sqrt{N} \quad (1.27)$$

hence from Equation (1.26) we have:

$$\sum_{i=1}^K \sum_{t=1}^{n_T(i)} \sqrt{\frac{1}{t}} \leq 2 \sum_{i=1}^K \sqrt{n_T(i)}. \quad (1.28)$$

Since all $n_T(i)$ sum to T , by the Jensen's inequality (concavity of $\sqrt{\cdot}$) we have:

$$2 \sum_{i=1}^K \sqrt{n_T(i)} = 2K \left(\frac{1}{K} \sum_{i=1}^K \sqrt{n_T(i)} \right) \leq 2K \sqrt{\frac{1}{K} \sum_{i=1}^K n_T(i)} = 2K \sqrt{\frac{T}{K}} = 2\sqrt{KT}. \quad (1.29)$$

■

1.2 Contextual and Bayesian Bandits

We introduce the *context*: there is a set X of contexts/states, $x_t \in X$ is the state at time t , the reward depends not only on the action but also on the current state: $r(x_t, a_t)$. For simplicity we assume that the reward function is deterministic.

- At each time step, we are in the state x_t .
- We choose an action a_t and observe $r(x_t, a_t)$.
- The next state x_{t+1} does not depend on the action, but is sampled from a distribution μ .

A **policy** π is a mapping from *all* the states to the actions. Is important to emphasize that a policy must assign an action to **each possible state**; if it is not defined on a particular state, then it is not a policy.

- Can be deterministic: $a = \pi(s)$,
- or stochastic $\pi(a | s)$.

Note how we denoted s a state, this is a general notation. In the bandits context, we will use x to denote the state. We define the **optimal policy** the one that maximizes the expected value for the reward:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{x \sim \mu} r(x, \pi(x)). \quad (1.30)$$

If the set of states X is discrete the expected value is

$$\pi^* = \arg \max_{\pi} \sum_{x \in X} r(x, \pi(x)) \mu(x) \quad (1.31)$$

with $\mu(x)^1$ being the probability of sampling the state x .

¹Formally, we should write $\mu(\{x\})$, since μ is a measure on X .

Let $\{\pi^0, \pi^1, \dots, \pi^{T-1}\}$ be the sequence of policies that we learned at each time step from the beginning to $t = T - 1$. At the T -th iteration, we define the **regret** as the total expected reward obtained using the optimal policy π^* minus the total expected reward obtained by using our sequence of policies:

$$\text{Regret}_T = T \mathbb{E}_{x \sim \mu} [r(x, \pi^*(x))] - \sum_{t=0}^{T-1} \mathbb{E}_{x \sim \mu} [r(x, \pi^t(x))] \quad (1.32)$$

we recall that at every time step, the states are sampled by the distribution μ independently from the previous states.

1.2.1 Explore and Commit

Let's see how the explore & commit algorithm works on the contextual bandit, the goal is to build a *table* that assign to each pair of state and action, the estimated expected value of the reward.

Algorithm 1 Explore & Commit

- 1: **for** $t = 0, \dots, N - 1$ (**explore**) **do**
- 2: observe the current state $x_t \sim \mu$
- 3: sample a random action from a uniform distribution $a_t \sim \text{Unif}(A)$
- 4: observe the reward $r_t = r(x_t, a_t)$
- 5: build an unbiased estimator $\mathbf{r}_t$ defined as follows:

$$\mathbf{r}_t[a] = \begin{cases} 0 & \text{if } a \neq a_t \\ r_t \cdot |A| & \text{else} \end{cases} \quad (1.33)$$

at each time step t the estimator assigns zero to all the actions $\mathbf{r}_t$ except for the one used at time t , to which it assigns the reward multiplied by the number of actions.

- 6: **end for**
- 7: compute the policy as follows:

$$\hat{\pi} = \arg \max_{\pi} \sum_{i=0}^{N-1} \mathbf{r}_i[\pi(x_i)] \quad (1.34)$$

- 8: **for** $t = N, \dots, T - 1$ (**commit**) **do**
 - 9: observe the current state $x_t \sim \mu$
 - 10: play the arm $a_t = \hat{\pi}(x_t)$
 - 11: **end for**
-

About the estimator: We want to give a description of how the estimator $\mathbf{r}_t$ works. There are N vectors $\mathbf{r}_0, \mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{N-1}$, each of dimension $|A|$ (the number of possible actions). For a given time step t , $\mathbf{r}_t$ is null on each action, except for the action chosen at time t :

$$\mathbf{r}_0 = \begin{pmatrix} r_0/p(a_0) \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{r}_1 = \begin{pmatrix} 0 \\ r_1/p(a_1) \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{r}_t = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ r_t/p(a_t) \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{r}_{N-1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \vdots \\ r_{N-1}/p(a_{N-1}) \end{pmatrix}. \quad (1.35)$$

$p(a_t)$ is the probability of choosing the action a_t , since they are sampled from a uniform distribution, for each action a we have $p(a) = \frac{1}{|A|}$ hence $\frac{r_t}{p(a_t)} = r_t \cdot |A|$. The regret for this algorithm has the following asymptotic behavior:

$$O(T^{2/3} \cdot |A|^{1/3} \cdot \ln^{1/3}(|\Pi|)) \quad (1.36)$$

where Π is the set of all the policies.

1.2.2 ϵ -Greedy

The ϵ -greedy algorithm never stops exploring, at each time step, with a given probability it selects a random action, else it selects the best action, exploiting the known information. We denote π^t the current policy that we are selecting as our best policy at time t . Let $\epsilon \in [0, 1]$, if $\epsilon = 0$, we just exploit the available information, if $\epsilon = 1$, we explore by randomly choosing the actions.

Algorithm 2 ϵ -Greedy

```

1: for  $t = 0, \dots, T - 1$  do
2:   observe the current state  $x_t \sim \mu$ 
3:   select a random number  $c \sim \text{Unif}([0, 1])$ 
4:   if  $c \leq \epsilon$  then
5:     sample a random action from a uniform distribution  $a_t \sim \text{Unif}(A)$  (explore)
6:   else
7:     choose the best action  $a_t = \pi^t(x_t)$  (exploit)
8:   end if
9:   observe the reward  $r_t = r(x_t, a_t)$ 
10:  build an unbiased estimator  $\mathbf{r}_t$  defined as follows:

```

$$\mathbf{r}_t[a] = \begin{cases} 0 & \text{if } a \neq a_t \\ \frac{r_t}{p_t(a_t)} & \text{else} \end{cases} \quad (1.37)$$

where

$$p_t(a) = \frac{\epsilon}{|A|} + (1 - \epsilon)\mathbb{1}\{a = \pi^t(x_t)\} \quad (1.38)$$

```

11:   update the policy

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$$\pi^{t+1} = \arg \max_{\pi} \sum_{i=0}^t \mathbf{r}_i[\pi(x_i)] \quad (1.39)$$

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12: end for

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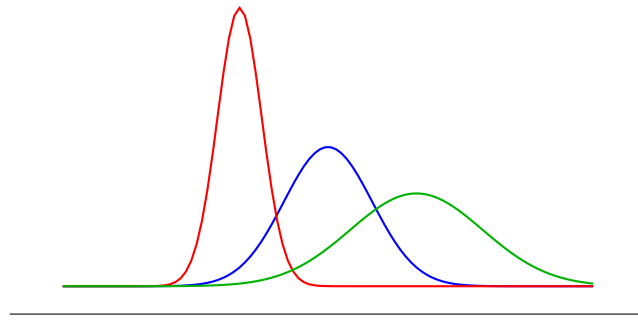
ϵ can also be time-varying, by making it high at the beginning (first explore) and then decaying as the time goes on (then exploit), for example $\epsilon(t) = 0.9 \exp(-t) + 0.1$.

1.2.3 Gaussian Bayesian Bandits

In the previous discussion, we treated the reward r as deterministic. Now let r be random again. We denote $\nu(i)$ the reward distribution, for the specific action a_i (hence we have a reward distribution for each action), we are making a step back by removing the context, the following discussion assumes that there is only one state. We exploit prior knowledge by *assuming* that the distributions $\nu(i)$ are **Gaussians**:

$$\nu(i) = \mathcal{N}(\mu(i), \sigma^2(i)) \quad (1.40)$$

with $\mu(i)$ and $\sigma^2(i)$ being the mean and variance for the reward by choosing the action a_i .



Usually, at the beginning we assume as prior knowledge that the mean of all the Gaussians is zero and the standard deviation is 1. In the algorithm we will start exploring and trying different actions,

sampling the rewards in order to fit the Gaussians with the correct values for the mean and the standard deviation.

Assume that we have fitted the Gaussians, we can apply the UCB to this case, the best action is the one with the highest upper bound for the reward, hence we choose the action with the Gaussian with the highest standard deviation (considering also the mean). Let K be the number of actions:

$$a_t = \arg \max_{1 \leq i \leq K} \mu_t(i) + \frac{c\sigma_t(i)}{\sqrt{N_t(i)}} \quad (1.41)$$

where

- $\mu_t(i), \sigma_t(i)$ are the current estimation at time t for the mean and the standard deviation of the distributions of the i -th action.
- c is an exploration parameter.
- $N_t(i)$ is the total number of times that the i -th action has been selected.

CHAPTER

2

MARKOVIAN PROCESSES

2.1 Markov Decision Processes

First of all we introduce a *sequential* decision making approach. We see the agent that has to learn the policy as an entity that can interact with the world/environment through actions a_t , and can receive from the environment an observation o_t and a reward r_t after each action. The agent interacts at discrete time steps; such discrete interaction generates a **trajectory** (or history) of tuples (observation-action-reward):

$$\underbrace{h_t}_{\text{history at time } t} = (o_0, a_0, r_1, o_1, a_1, r_2, \dots, r_t, o_t, a_t). \quad (2.1)$$

We define a state function $s_t = f(h_t)$ that describes the current state of the agent at time t and depends on the history h_t and is typically hidden or unknown.

Definition 2.1.1 : Markov Assumption

A state s_t is **Markovian** if the future is independent of the past information, given the present state: the probability of ending up in a given state s_{t+1} by applying a_t at time t , depends only on the current state s_t and not on the history h_t :

$$\mathbb{P}(s_{t+1} \mid s_t, a_t) = \mathbb{P}(s_{t+1} \mid h_t, a_t). \quad (2.2)$$

A state can always be made Markovian if we define the state exactly as the previous history:

$$s_t = h_t. \quad (2.3)$$

In practice the current state is defined as the *latest observation*:

$$s_t = o_t \quad (2.4)$$

in this case the state is said to be **fully observable**.

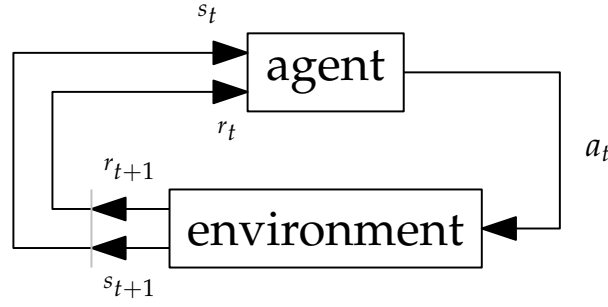


Figure 2.1: Markov Decision Process scheme.

A **Markov Decision Process (MDP)** is a sequential decision making scheme under the Markov assumption. The transition dynamics is generally stochastic: $\mathbb{P}(s_{t+1} \mid s_t, a_t)$. An alternative notation to denote the fact that the next state is stochastic by applying a_t on s_t is the following:

$$s_{t+1} \sim \mathbb{P}(\cdot \mid s_t, a_t) \quad (2.5)$$

$$\text{or } s' \sim \mathbb{P}(\cdot \mid s, a) \quad (2.6)$$

A **reward** r_t is a scalar signal representing feedback on the actions taken by the agent, and indicates how well an agent is doing at a given time step t . The reward is a function of the state and action (reward taken by applying a given action at a given state):

$$r_{t+1} = R(s_t, a_t). \quad (2.7)$$

We define the **cost** as the opposite of the reward. Although in reinforcement learning the objective is always to maximize the reward function, the fact that this is the best way to reach a goal is not guaranteed, in fact the so-called **Reward Hypothesis**:

“all goals can be described as the maximization of expected cumulative reward.”

remains an open question today.

Definition 2.1.2 : Markov Decision Process

An MDP is defined by a 5-tuple:

$$(S, A, \mathbb{P}, R, \gamma) \quad (2.8)$$

where:

- S is the set of all possible states.
- A is the set of all the actions the agent can take.
- $\mathbb{P}$ is the transition probability, which defines the dynamics of the environment.
- R is the reward function.
- γ is the **discount factor**, a real value within the interval $[0, 1)$ that determines the importance of future rewards compared to immediate ones.

The behavior of an agent is given by a policy, a function that determines how agents select actions.

Definition 2.1.3 : Policy

A **policy** is a function $\pi : S \rightarrow \Delta(A)$ that assigns a distribution over the set of actions to every state.

A policy can be deterministic (special case where $\pi : S \rightarrow A$) and we write:

$$\pi(s) = a \quad (2.9)$$

or stochastic:

$$\pi(a | s) \in [0, 1] \quad \text{with} \quad \sum_{a \in A} \pi(a | s) = 1. \quad (2.10)$$

We can also write $a \sim \pi(\cdot | s)$. Let's define the **probability of a trajectory**: What's the probability of seeing a given trajectory at time t according to π starting at s_0 ? We have the trajectory

$$(s_0, a_0, s_1, \dots, s_t, a_t) \quad (2.11)$$

and

$$\mathbb{P}^\pi(a_0, s_1, \dots, s_t, a_t | s_0) \quad (2.12)$$

$$= \pi(a_0 | s_0) \mathbb{P}(s_1 | s_0, a_0) \pi(a_1 | s_1) \mathbb{P}(s_2 | s_1, a_1) \dots \mathbb{P}(s_t | s_{t-1}, a_{t-1}) \pi(a_t | s_t) \quad (2.13)$$

notice how we added π as superscript to denote the fact that this probability depends on the given policy function. We can also define the **state visitation probability**, i.e. the probability of visiting a particular state $\hat{s}$ at time t according to π starting from s_0 :

$$\mathbb{P}_t^\pi(s_t = \hat{s} | s_0) = \sum_{a_0, s_1, \dots, a_{t-1}} \mathbb{P}^\pi(s_0, a_0, s_1, \dots, s_t = \hat{s}). \quad (2.14)$$

The summation in equation (2.14) sums over all trajectories of length t that start at s_0 and end in $\hat{s}$.

A basic example of a Markov decision process is a robotic manipulator with a hand gripper that can grab and carry a ball:

- the states are the possible robot configurations (joint values) and the ball position,
- the actions are the possible torques on the arm and the finger joints,
- the transition is stochastic, given by the physics (how the torques update the joint values over time) and by the noise,
- the policy maps each configuration+ball position to a torque for the joints,
- the cost (opposite of the reward) is the magnitude of the torque and the distance to the goal.

2.1.1 MDP Horizon

In Definition 2.1.2 we introduced a discount factor $\gamma \in [0, 1]$. This term weights the importance of immediate rewards with respect to future rewards:

- $\gamma = 0$: Our agent only cares about immediate rewards.
- $\gamma = 1$: Immediate and future rewards are equally important.

The discount factor is used in the **Value Function** V^π , a function that estimates the goodness of a state based on its value. It is also a measure to compare different policies.

Definition 2.1.4 : Value Function

The **Value Function** V^π measures the expected cumulative reward (the sum of all future rewards, possibly discounted by the γ factor) starting from s and continuing to follow the policy π for all subsequent actions.

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{h=0}^{\infty} \gamma^h r_{h+1} \mid s_0 = s \right] \quad (2.15)$$

$$= \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h \cdot \underbrace{r_{h+1}}_{R(s_h, a_h)} \mid s_0 = s, a_h \sim \pi(\cdot | s_h), s_{h+1} \sim \mathbb{P}(\cdot | s_h, a_h) \right]. \quad (2.16)$$

With $\mathbb{E}$ we denote the expected value. The value function V^π considers all the possible infinite trajectories that are generated by applying π starting from the argument s , since the transition and the policy are (or might be) stochastic, a probability is assigned to each trajectory, so the expected value is considered, i.e. the sum over the cumulative reward of a given trajectory, weighted by the probability of that trajectory.

Example: Let's assume that there is an MDP with three possible trajectories starting from s_a :

$$\text{trajectory 1 : } (s_a, a, s_c) \quad (2.17)$$

$$\text{trajectory 2 : } (s_a, a, s_b, a, s_c) \quad (2.18)$$

$$\text{trajectory 3 : } (s_a, a, s_b, a, s_b, a, s_c) \quad (2.19)$$

There is only one action (always chosen by the policy) and the reward function is deterministic:

$$R(s_a, a) = 2 \quad (2.20)$$

$$R(s_b, a) = 1 \quad (2.21)$$

$$R(s_c, a) = 0 \quad (2.22)$$

Usually these trajectories are infinite, but in this case we assume that they all end up in s_c , an ending state where all actions have a reward of zero with probability 1:

$$\mathbb{P}(s_c | s_c, a) = 1. \quad (2.23)$$

The expected value of the cumulative reward assuming the following probabilities:

$$\mathbb{P}(s_a, a, s_c) = 0.2 \quad (2.24)$$

$$\mathbb{P}(s_a, a, s_b, a, s_c) = 0.1 \quad (2.25)$$

$$\mathbb{P}(s_a, a, s_b, a, s_b, a, s_c) = 0.7 \quad (2.26)$$

assuming $\gamma = 1$ is

$$2 \cdot 0.2 + 3 \cdot 0.1 + 4 \cdot 0.7 = 3.5. \quad (2.27)$$

The Action-Value function Q^π goes one step further than V^π , tying the value not only to the state, but also to the action you choose to take.

Definition 2.1.5 : Action-Value Function

The **Action-Value Function** Q^π measures the expected value of the cumulative reward starting from a state s , performing a specific action a and then continuing to follow the policy π .

$$Q^\pi(s, a) = \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h \cdot \underbrace{r_{h+1}}_{R(s_h, a_h)} \mid (s_0, a_0) = (s, a), \underbrace{a_h \sim \pi(\cdot \mid s_h)}_{\forall h \geq 1}, s_{h+1} \sim \mathbb{P}(\cdot \mid s_h, a_h) \right]. \quad (2.28)$$

Theorem 2.1.1 : The Bellman Equation

The value of a certain state can be expanded in terms of the current reward and the value of the next states according to the policy:

$$V^\pi(s) = \sum_{a \in A} \pi(a \mid s) \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' \mid s, a) V^\pi(s') \right]. \quad (2.29)$$

The Bellman equation can be applied to the action-value function:

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' \mid s, a) V^\pi(s'). \quad (2.30)$$

In Equation (2.29) V^π is a function of s and $\pi(s)$ while in equation (2.30) Q^π is a function of s and a (as highlighted by the green characters). As a result of these equations we have the following identity:

$$V^\pi(s) = \sum_{a \in A} \pi(a | s) Q^\pi(s, a) \quad (2.31)$$

if π is deterministic:

$$V^\pi(s) = Q^\pi(s, \pi(s)) \quad (2.32)$$

V^π can be expressed in terms of Q^π . One last quantity that we introduce is the following.

Definition 2.1.6 : Discounted State-Action Distribution

The discounted state-action distribution quantifies how often (and with what discounted time weight) the pair (s, a) is visited along the entire infinite horizon.

$$d_{s_0}^\pi(s, a) = (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h \mathbb{P}^\pi(s_h = s | s_0) \pi(a | s). \quad (2.33)$$

Since there is a discount factor $\gamma < 1$, events that occur far away in time matter less and less.

$\mathbb{P}^\pi(s_h = s | s_0) \pi(a | s)$ is the exact probability of being in state s and executing action a exactly at time step h , conditioned on having started in s_0 and having followed the π policy for all previous steps.

2.2 Optimality of a Policy

For an infinite horizon MDP there always exists a deterministic **optimal policy** π^* that satisfies:

$$V^{\pi^*}(s) \geq V^\pi(s), \quad \forall s, \forall \pi \quad (2.34)$$

This means that the policy π^* is the one that guarantees the greatest expected reward, since it dominates all other policies in each state. We will use also the following notation:

$$V^{\pi^*} = V^* \quad (2.35)$$

$$Q^{\pi^*} = Q^*. \quad (2.36)$$

The **Bellman Optimality** condition states that:

$$V^*(s) = \max_{a \in A} Q^*(s, a). \quad (2.37)$$

Example: Consider the MDP in Figure 2.2.

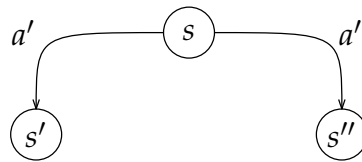


Figure 2.2: Three-states MDP example.

We assume that V^* is known in the states s' and s'' . We want to try a' and a'' in s in order to get the rewards and compute Q^* in s for the actions:

- We apply a' and get $R(s, a')$. We can compute $Q^*(s, a') = R(s, a') + \underbrace{\gamma V^*(s')}_{\text{known}}$.
- We apply a'' and get $R(s, a'')$. We can compute $Q^*(s, a'') = R(s, a'') + \underbrace{\gamma V^*(s'')}_{\text{known}}$.

Now we can use the identity in Equation (2.37) to compute $V^*(s)$:

$$V^*(s) = \max\{Q^*(s, a'), Q^*(s, a'')\}. \quad (2.38)$$

Theorem 2.2.1 : Bellman Optimality (Existence)

Let $\gamma \in [0, 1)$. The optimal value function $V^*(s)$ satisfies the nonlinear Bellman equation:

$$V^*(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right] \quad (2.39)$$

hence the policy:

$$\hat{\pi}(s) = \arg \max_{a \in A} Q^*(s, a) \quad (2.40)$$

is the optimal policy: $V^{\hat{\pi}} = V^*$.

Proof: First we prove that V^* satisfies the Bellman Equation (2.39). For a generic policy π , we can write the Bellman equation by isolating the first action $a \sim \pi(\cdot | s)$:

$$V^\pi(s) = \sum_{a \in A} \pi(a | s) \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^\pi(s') \right]. \quad (2.41)$$

The optimal value function can be written as follows:

$$V^*(s) = \max_{\pi} \sum_{a \in A} \pi(a | s) \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^\pi(s') \right]. \quad (2.42)$$

Consequently, the maximum over π coincides with the maximization over a single action $a \in A$ by placing $V^*(s')$ in the future steps:

$$V^*(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right] \quad (2.43)$$

hence V^* satisfies the Bellman equation. Now we have to prove that $\hat{\pi}$ is the optimal policy. We define the deterministic policy $\hat{\pi}$ as in Equation (2.40). Since $\hat{\pi}(s)$ selects the action that maximizes Q^* we can write:

$$V^*(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right] \quad (2.44)$$

$$= R(s, \hat{\pi}(s)) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) V^*(s'). \quad (2.45)$$

On the other hand, from the Bellman expectation equation applied to $\hat{\pi}$ we can write:

$$V^{\hat{\pi}}(s) = R(s, \hat{\pi}(s)) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) V^{\hat{\pi}}(s') \quad (2.46)$$

if we subtract (2.46) from (2.45) we get:

$$V^*(s) - V^{\hat{\pi}}(s) = \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) (V^*(s') - V^{\hat{\pi}}(s')). \quad (2.47)$$

Now we take the absolute value of both sides:

$$|V^*(s) - V^{\hat{\pi}}(s)| = \left| \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) (V^*(s') - V^{\hat{\pi}}(s')) \right|. \quad (2.48)$$

By the triangle inequality:

$$|V^*(s) - V^{\hat{\pi}}(s)| = \left| \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) (V^*(s') - V^{\hat{\pi}}(s')) \right| \quad (2.49)$$

$$\leq \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) |V^*(s') - V^{\hat{\pi}}(s')| \quad (2.50)$$

$$\leq \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) \|V^* - V^{\hat{\pi}}\|_{\infty} \quad (2.51)$$

we used the L_∞ norm, the following inequality holds:

$$|V^*(s) - V^{\hat{\pi}}(s)| \leq \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) \|V^* - V^{\hat{\pi}}\|_\infty. \quad (2.52)$$

$\|V^* - V^{\hat{\pi}}\|_\infty$ is constant and $\sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) = 1$, hence

$$|V^*(s) - V^{\hat{\pi}}(s)| \leq \gamma \|V^* - V^{\hat{\pi}}\|_\infty. \quad (2.53)$$

this holds for each s hence:

$$\|V^* - V^{\hat{\pi}}\|_\infty \leq \gamma \|V^* - V^{\hat{\pi}}\|_\infty \implies \quad (2.54)$$

$$\|V^* - V^{\hat{\pi}}\|_\infty = 0 \implies \quad (2.55)$$

$$V^{\hat{\pi}}(s) = V^*(s), \quad \forall s. \quad (2.56)$$

■

Theorem 2.2.2 : Bellman Optimality (Uniqueness)

Let $\gamma \in [0, 1)$. For any value function V , if

$$V(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V(s') \right] \quad \forall s \in S \quad (2.57)$$

then $V(s) = V^*(s)$ for all $s \in S$.

Proof: Let V be any value function satisfying the Bellman optimality equation for all $s \in S$. Since V^* is also a solution, we can write (for any state s):

$$|V(s) - V^*(s)| \quad (2.58)$$

$$= \left| \max_{a \in A} \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V(s') \right) - \max_{a \in A} \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right) \right|. \quad (2.59)$$

Using the property $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$:

$$|V(s) - V^*(s)| \quad (2.60)$$

$$\leq \max_{a \in A} \left| \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V(s') \right) - \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right) \right| \quad (2.61)$$

we can cancel out $R(s, a)$ and move γ outside the max function:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \left| \sum_{s' \in S} \mathbb{P}(s' | s, a) (V(s') - V^*(s')) \right| \quad (2.62)$$

by the triangle inequality we can move the absolute value inside the sum:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \sum_{s' \in S} \mathbb{P}(s' | s, a) |V(s') - V^*(s')|. \quad (2.63)$$

We use the definition of L_∞ norm:

$$|V(s') - V^*(s')| \leq \max_{y \in S} |V(y) - V^*(y)| = \|V - V^*\|_\infty \quad (2.64)$$

Equation (2.63) becomes:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \sum_{s' \in S} \mathbb{P}(s' | s, a) \|V - V^*\|_\infty \quad (2.65)$$

$\|V - V^*\|_\infty$ is a constant hence can be moved out of the summation:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \left(\|V - V^*\|_\infty \sum_{s' \in S} \mathbb{P}(s' | s, a) \right) \quad (2.66)$$

trivially

$$\sum_{s' \in S} \mathbb{P}(s' | s, a) = 1$$

hence

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} (\|V - V^*\|_\infty \cdot 1) = \gamma \|V - V^*\|_\infty \quad (2.67)$$

since this inequality holds for each state $s \in S$, we can write:

$$\underbrace{\max_{s \in S} |V(s) - V^*(s)|}_{\|V - V^*\|_\infty} \leq \gamma \|V - V^*\|_\infty \implies \quad (2.68)$$

$$\|V - V^*\|_\infty \leq \gamma \|V - V^*\|_\infty \implies \quad (2.69)$$

$$(1 - \gamma) \|V - V^*\|_\infty \leq 0 \implies \|V - V^*\|_\infty = 0. \quad (2.70)$$

This holds since $\gamma \in [0, 1)$, hence $(1 - \gamma) > 0$. By definition:

$$\|V - V^*\|_\infty = 0 \implies \quad (2.71)$$

$$V(s) = V^*(s) \quad \forall s \in S. \quad (2.72)$$

■

This formulation reduces the multi-step problem to a one-step greedy choice plus the continuation value $V(s')$. By uniqueness of the fixed point, any function V satisfying this optimality equation coincides with V^* .